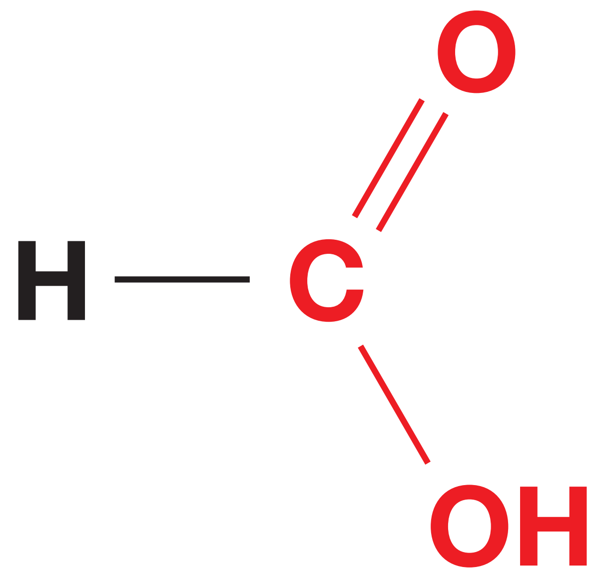
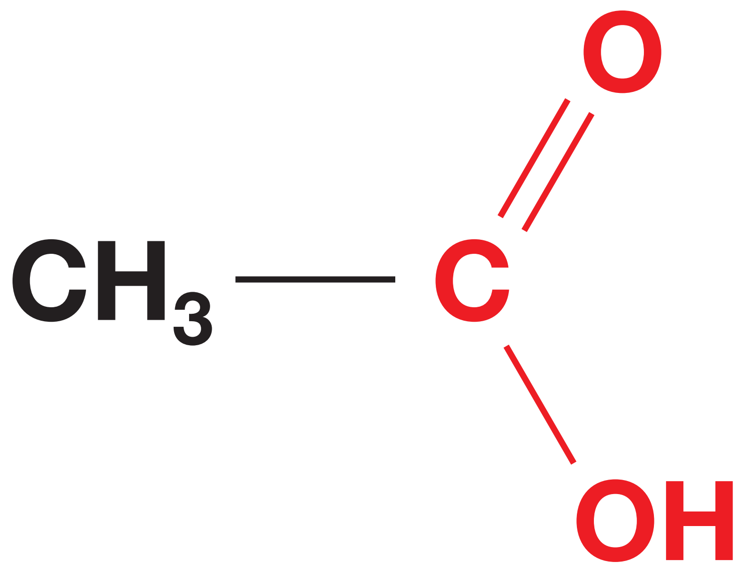
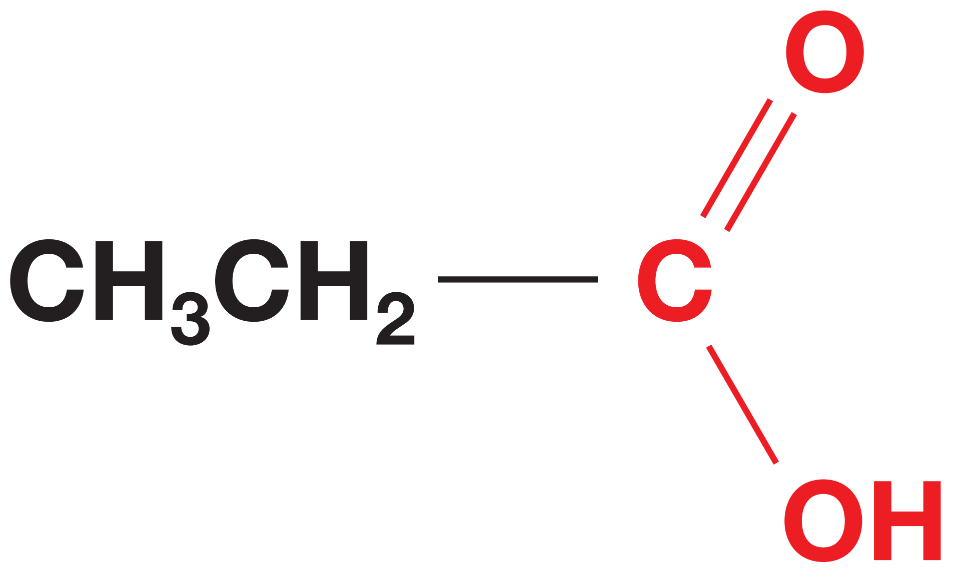
1 Carboxylic Acids: Names and Structures



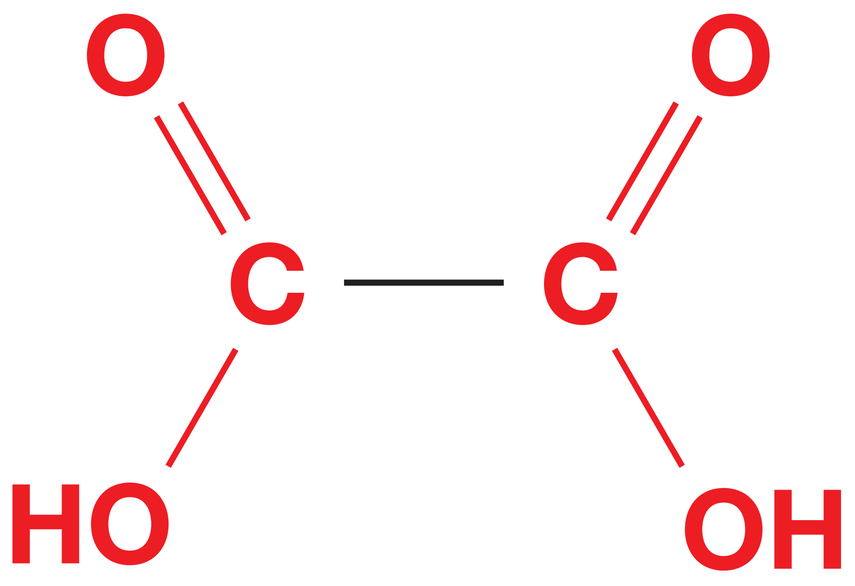
methanoic acid



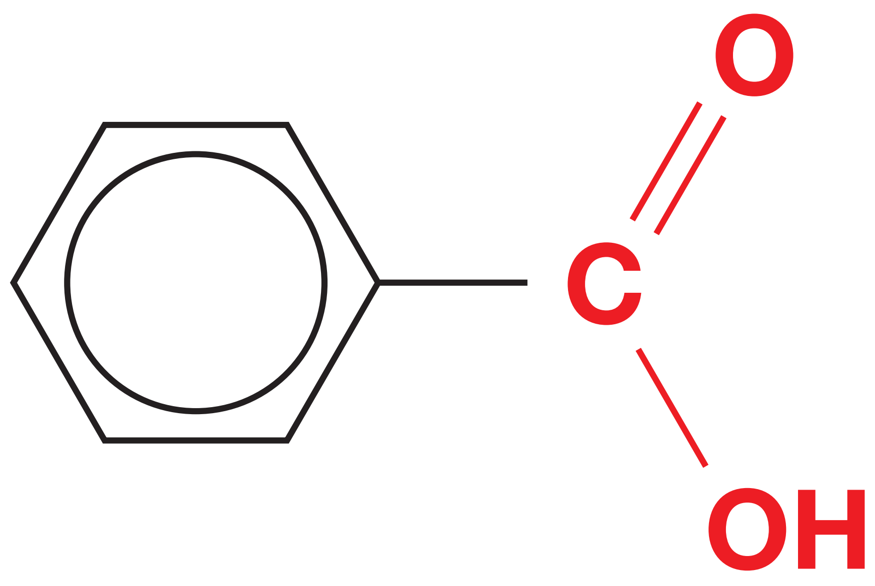
ethanoic acid



propanoic acid



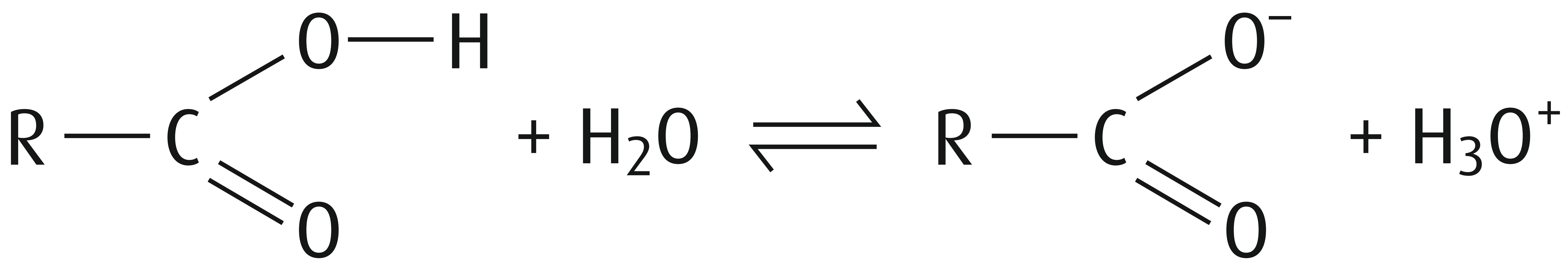
ethanedioic acid



benzoic acid

2 Acidity

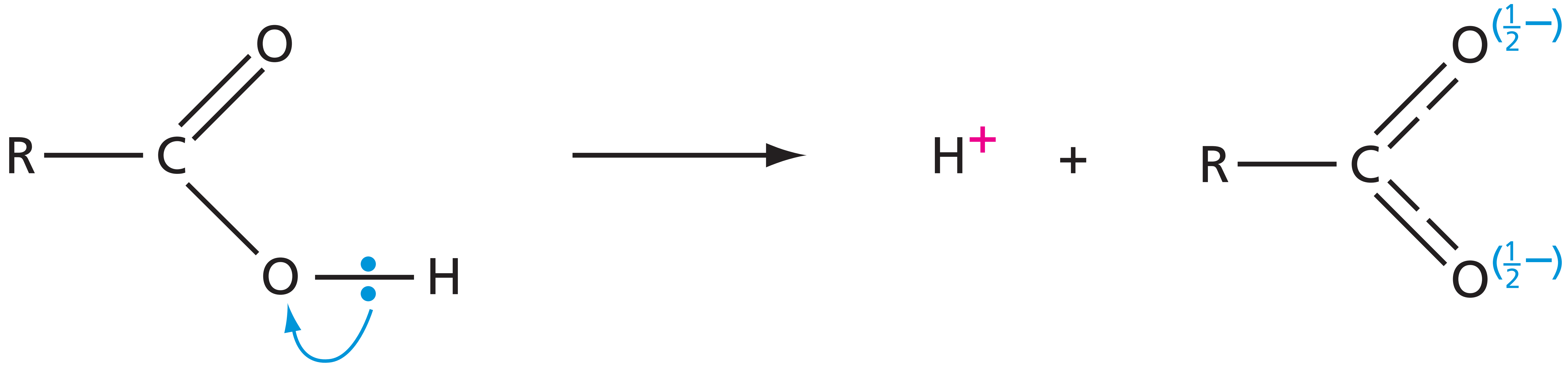
Carboxylic acids are weak acids that dissociate partially in aqueous solution according to the equation:

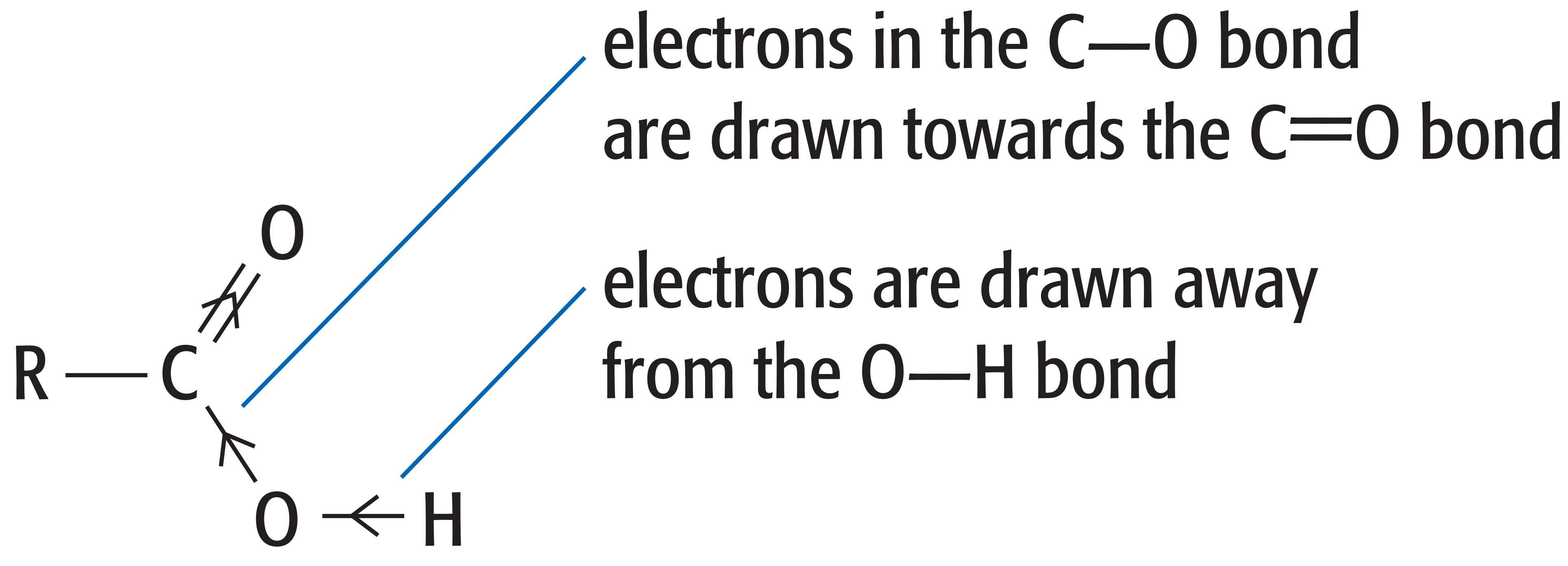


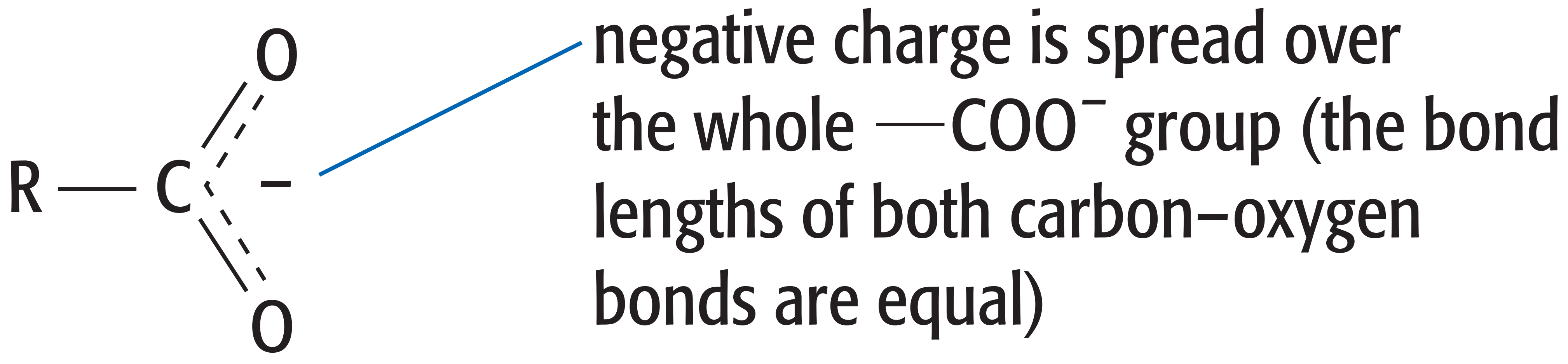
carboxylate ion

3 Acidity

They are more acidic than alcohols because the negative charge on the anion can be delocalised over two electronegative oxygen atoms.







4 Acidity

The increasing acidities of alcohols, phenols and carboxylic acids are explained by the increasing ability of the molecular structures to delocalise the negative charge in the alkoxide, phenoxide and carboxylate anions.

Their relative acidities are demonstrated by their different reactions with sodium, sodium hydroxide and sodium carbonate.

Reagent	**Observation with**		
	Hexanol	**Phenol**	**Hexanoic acid**
Na(s)	H_2(g) evolved	H_2(g) evolved	H_2(g) evolved
NaOH(aq)	no reaction	dissolves	dissolves
Na_2CO_3(aq)	no reaction	no reaction	CO_2(g) evolved

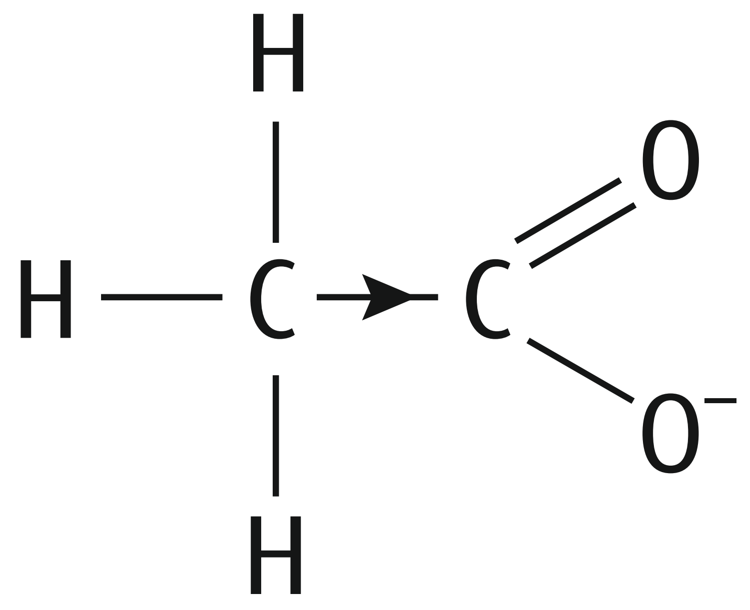
5 Relative Acidities

Atoms or groups that draw electrons away from the —CO_2^- group will help the anion to form, and this causes the acid to be more dissociated (that is, to become a stronger acid), e.g. halogens.

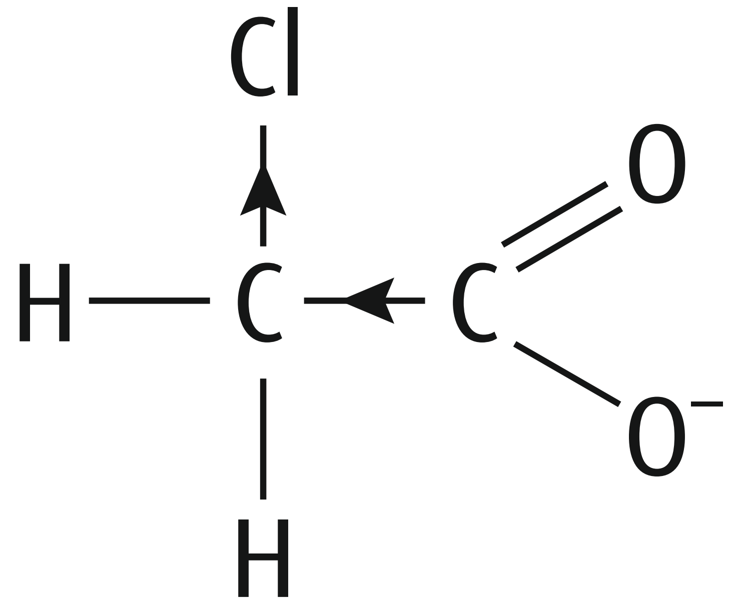
On the other hand, groups that donate electrons to the —CO_2^- group will cause the acid to become weaker, e.g. alkyl groups.

6 Relative Acidities

Electron-donating groups decrease the acid strength of carboxylic acids, whereas electron- withdrawing groups increase their acid strength.

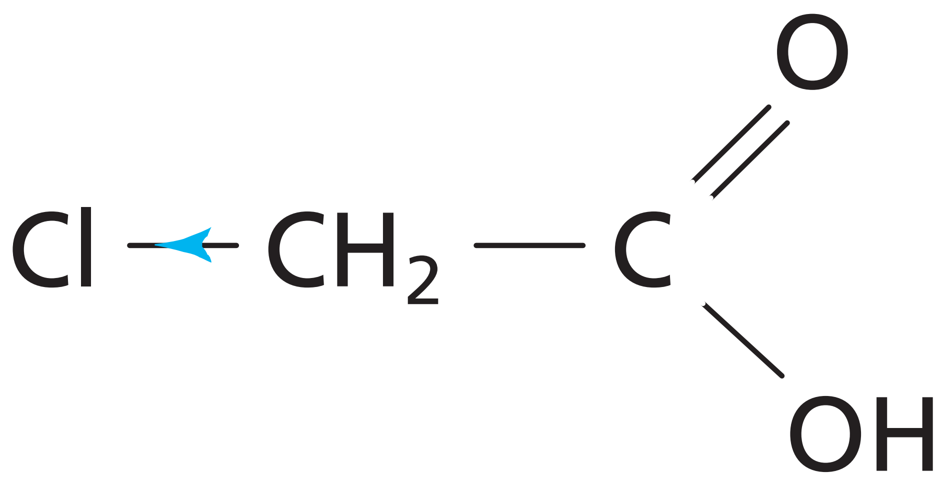
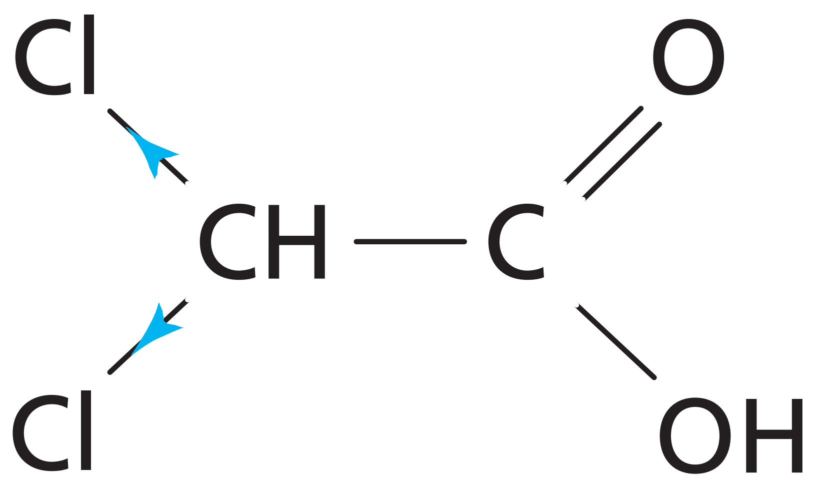
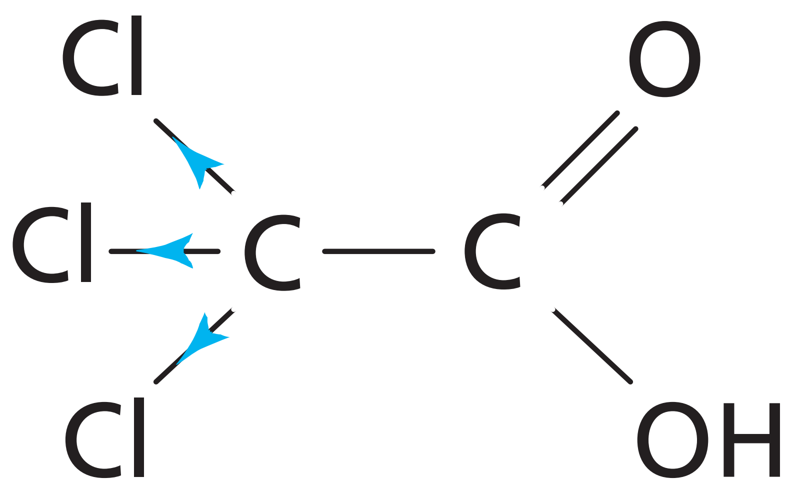


electron density donated to COO^- group



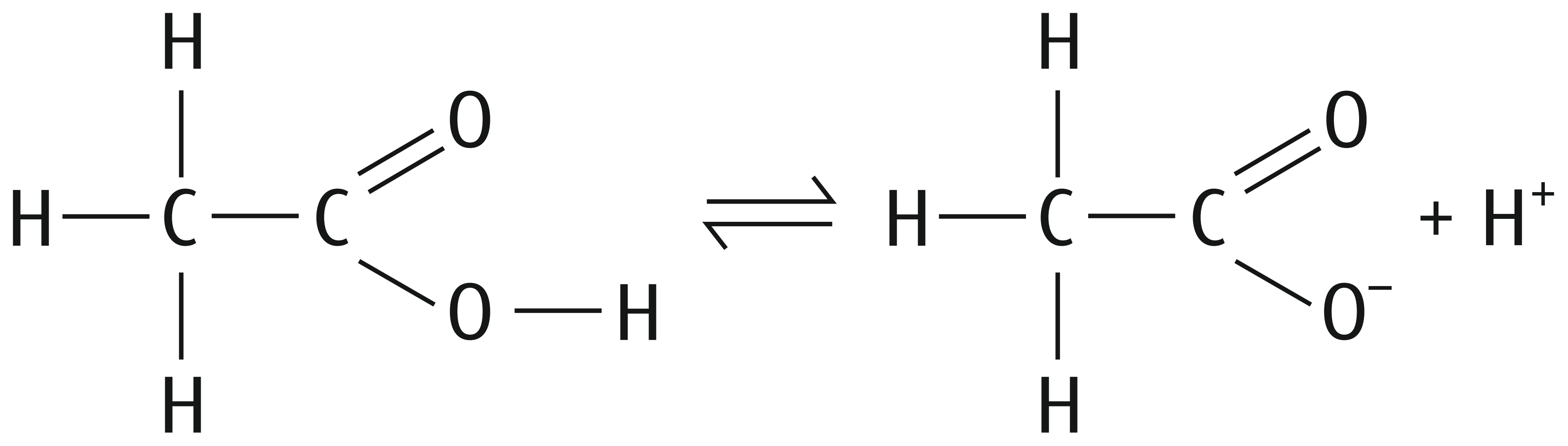
electron density withdrawn from COO^- group

7 Relative Acidities

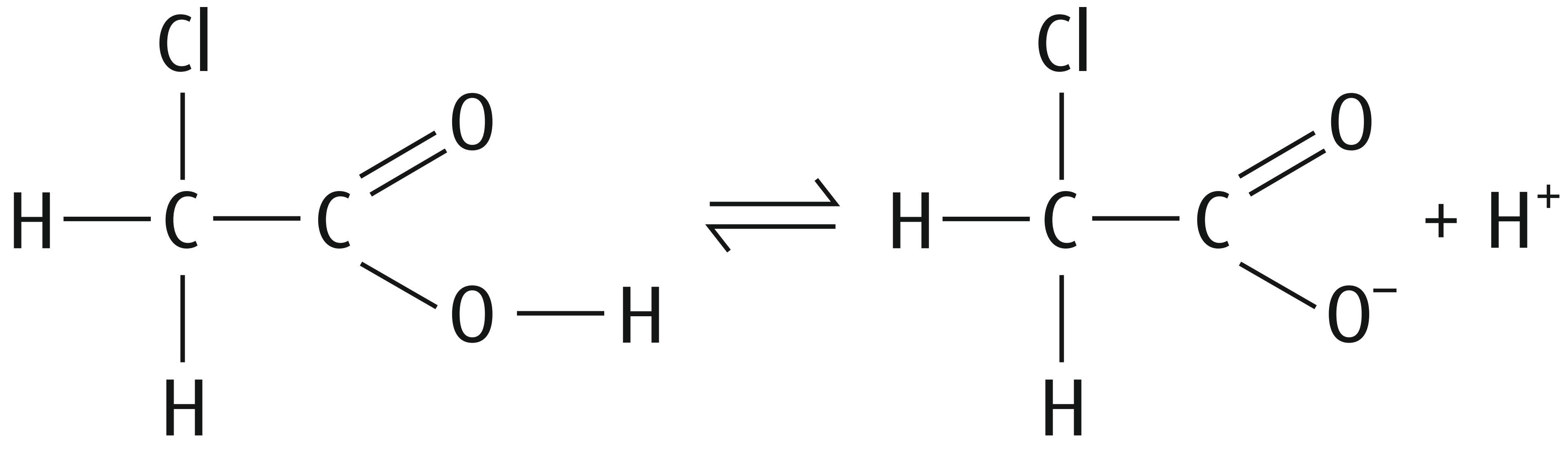
Formula of acid	pK_a	Percentage dissociation in $1.0\,mol\,dm^{-3}$ aqueous solution
	2.87	3.7%
	1.26	21%
	0.66	59%

8 Chloroethanoic Acid Vs Ethanoic Acid

Chloroethanoic acid is a stronger acid than ethanoic acid.



ethanoic acid ethanoate ion



chloroethanoic acid chloroethanoate ion

The difference in acidity can be explained by considering the stability of the anion (conjugate base) formed.

The more stable the anion (conjugate base) is, more likely is its formation.

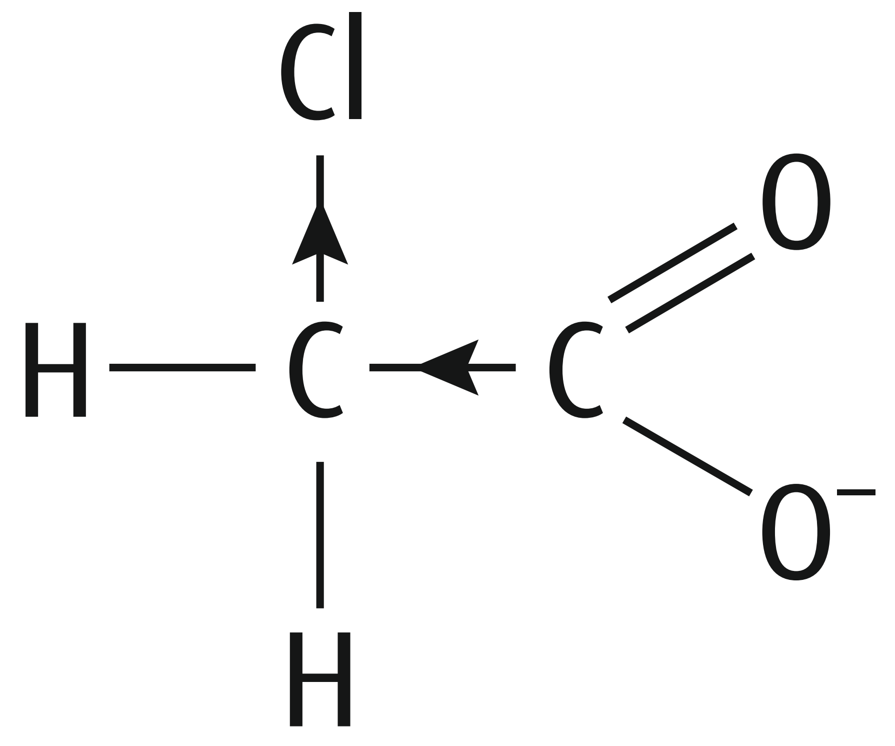
9 Chloroethanoic Acid Vs Ethanoic Acid

The Cl is a very electronegative atom and, in the chloroethanoate ion, pulls electron density away from the CO_2^- group.

Since the COO group is less negative, it attracts the H^+ ion back less strongly.

The conjugate base is, therefore, more stable.

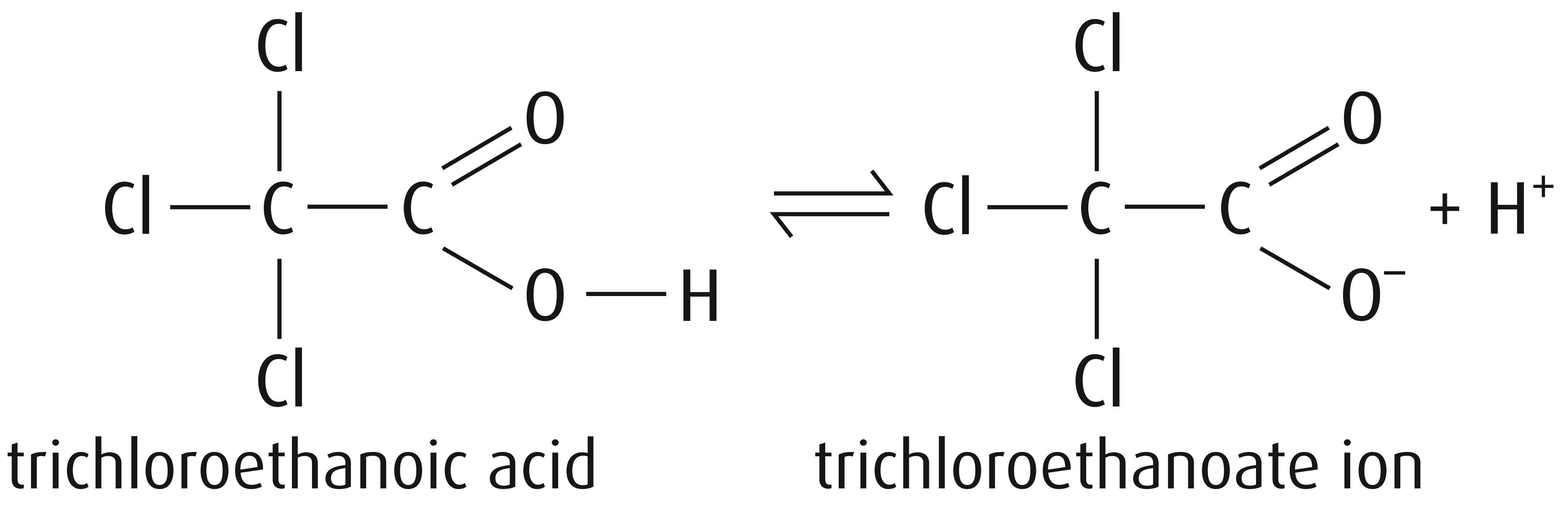
A more stable conjugate base indicates a greater tendency to dissociate and, therefore a stronger acid.

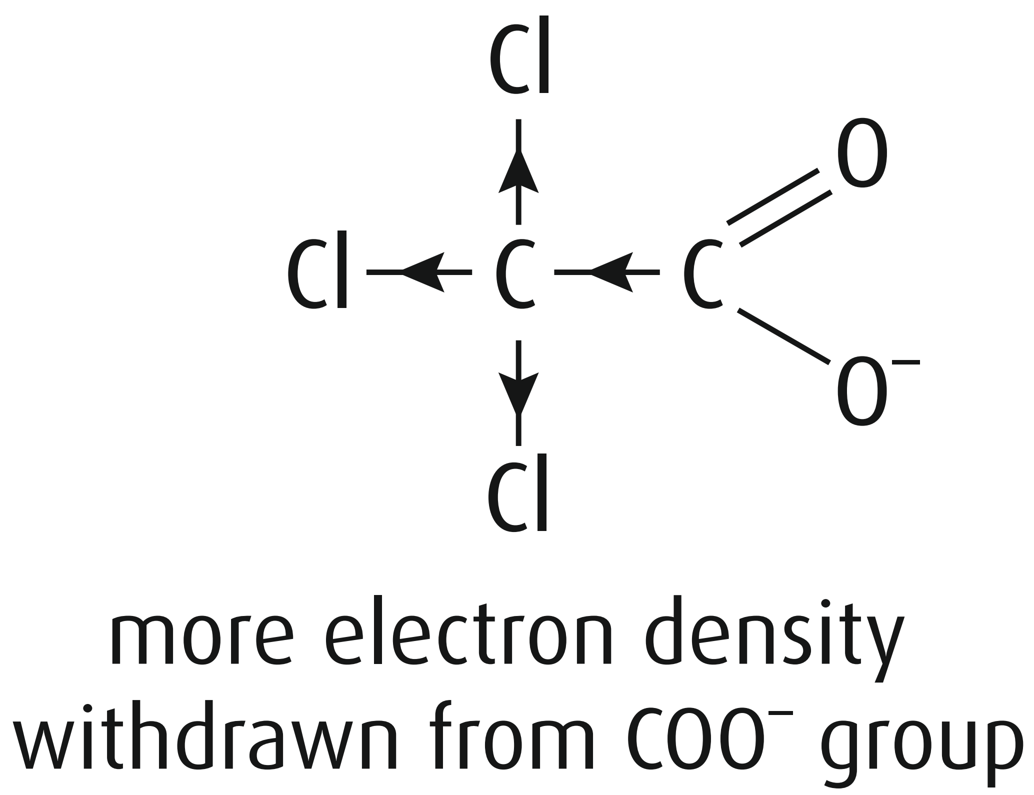


electron density withdrawn from COO^- group

10 Trichloroethanoic Acid Vs Chloroethanoic Acid

Trichloroethanoic acid is a stronger acid than chloroethanoic acid.





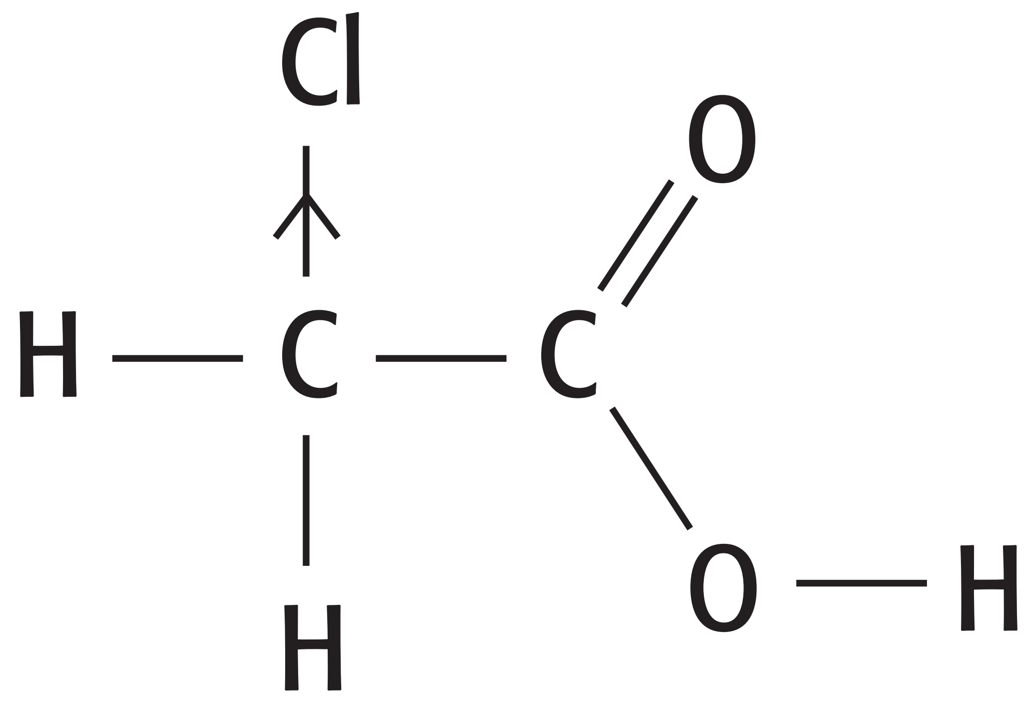
Three very electronegative Cl atoms have a greater electron-withdrawing effect than just one.

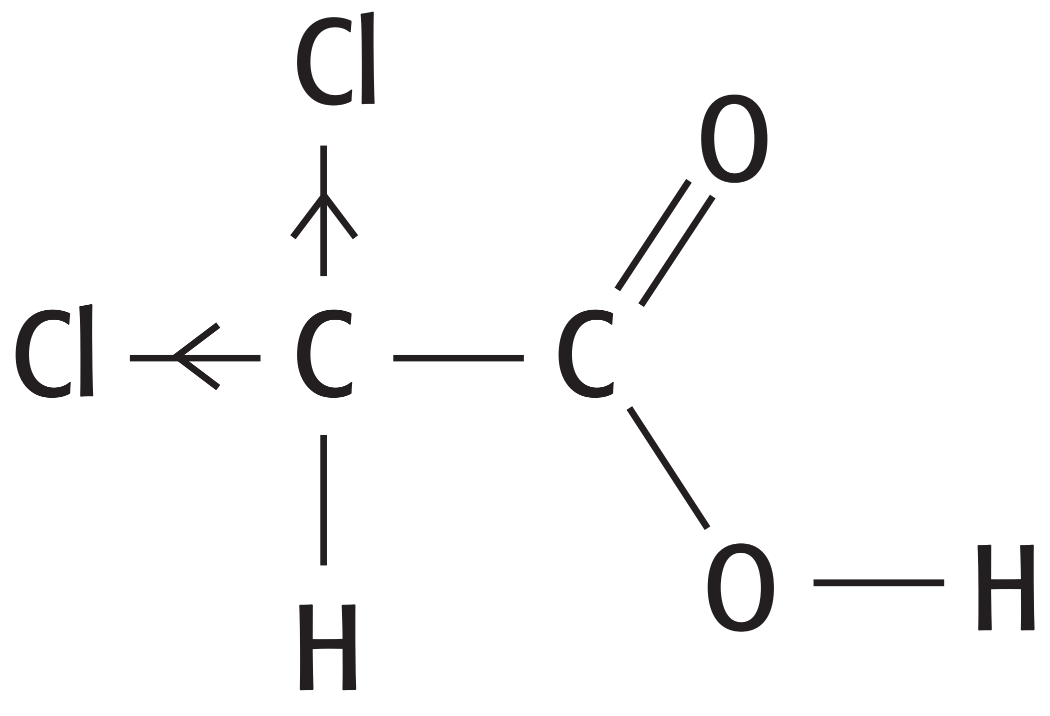
11 Trichloroethanoic Acid Vs Chloroethanoic Acid

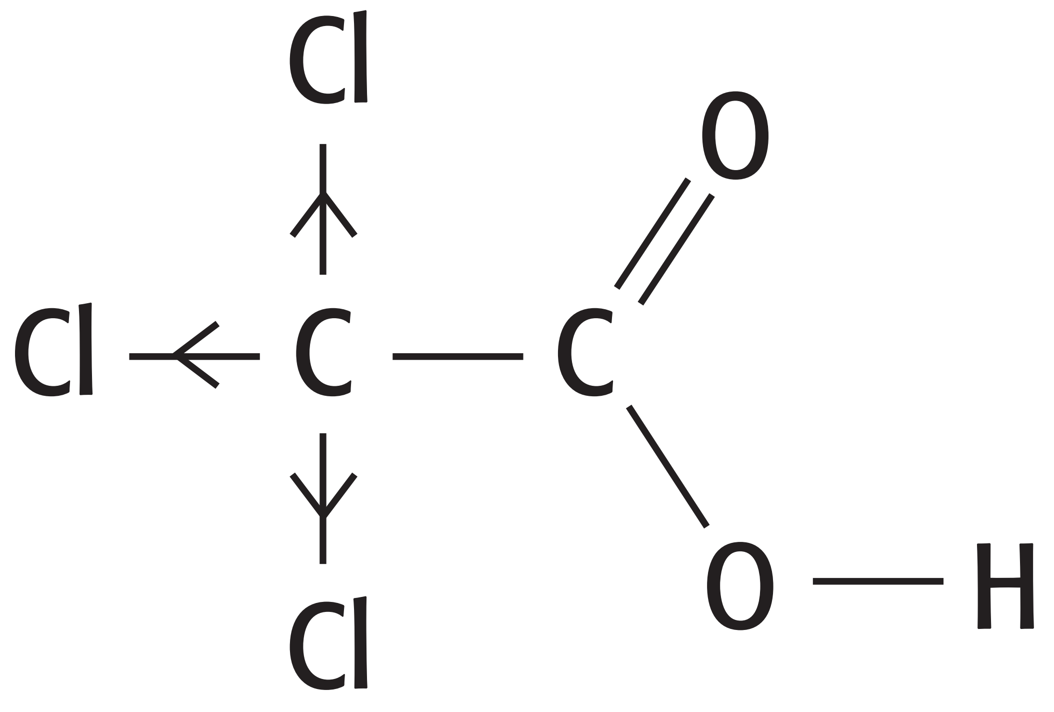
More electron density is thus pulled away from the COO– group making it less negative and less likely to attract back the H^+.

The trichloroethanoate ion is thus relatively more stable than the chloroethanoate ion and so trichloroethanoic acid dissociates more.

12 Multiple Chlorine Substitutes on Acids





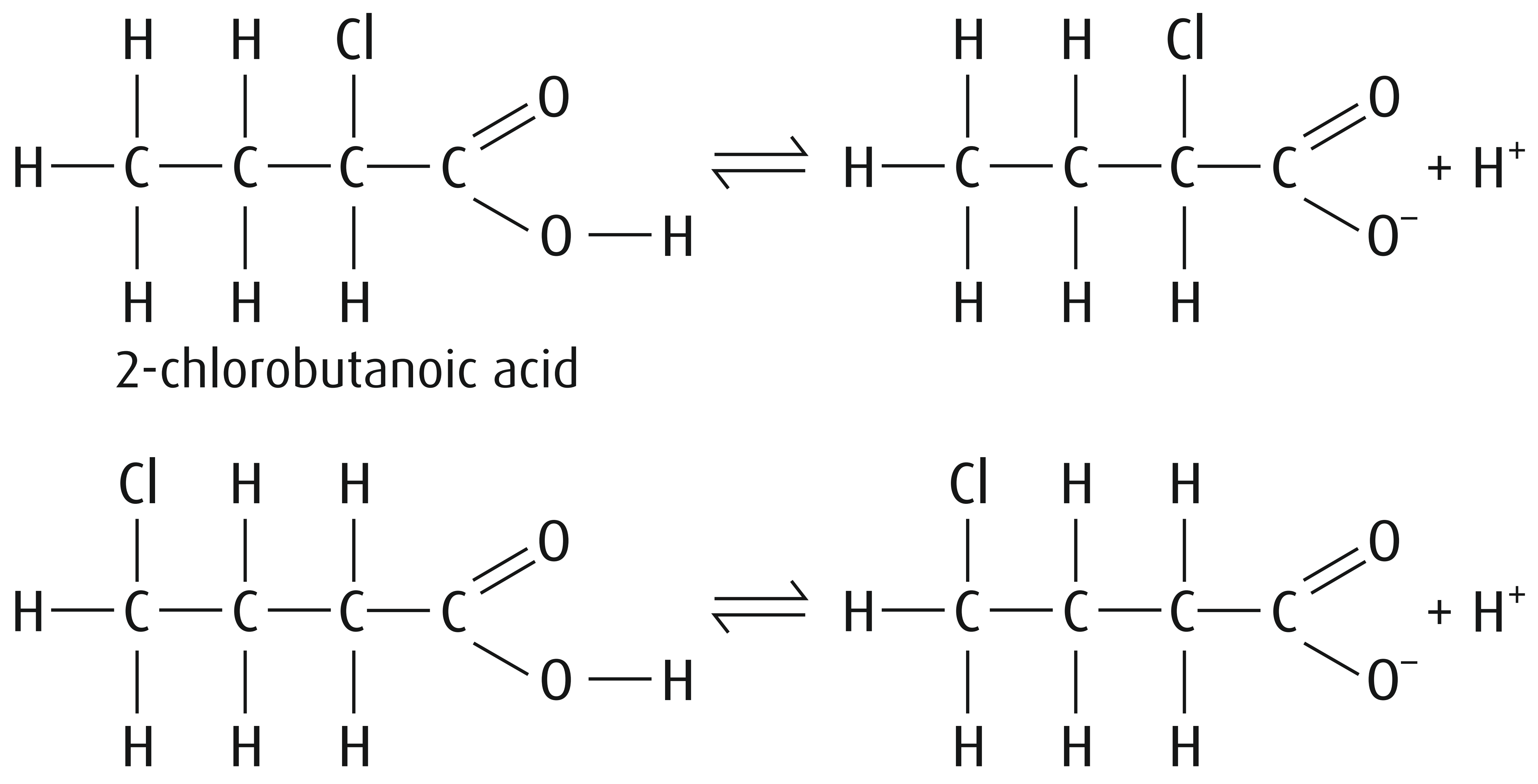


	$CH_2ClCOOH$	$CHCl_2COOH$	CCl_3COOH
pK_a	2.87	1.29	0.65

13 Effect of Distance of Cl Atoms on Acidity

2-Chlorobutanoic acid is a stronger acid than 4-chlorobutanoic acid.

The further the Cl is from the CO_2^- group in the anion, the smaller the electron-withdrawing effect it has on it.



2-chlorobutanoic acid

4-chlorobutanoic acid

14 Oxidation of Methanoic Acid

However, methanoic acid (HCO_2H) is can undergo oxidation.

This involves the oxidising agent breaking down the methanoic acid molecule, forming carbon dioxide.

This oxidation can be carried out even by warming with mild oxidising agents (such as the Fehling's or Tollens' reagent).

When methanoic acid is oxidised by Fehling's solution, tthe Cu^{2+} ion in Fehling's solution is reduced to the Cu^{+} ion, which precipitates out as red copper(I) oxide.

15 Oxidation of Methanoic Acid

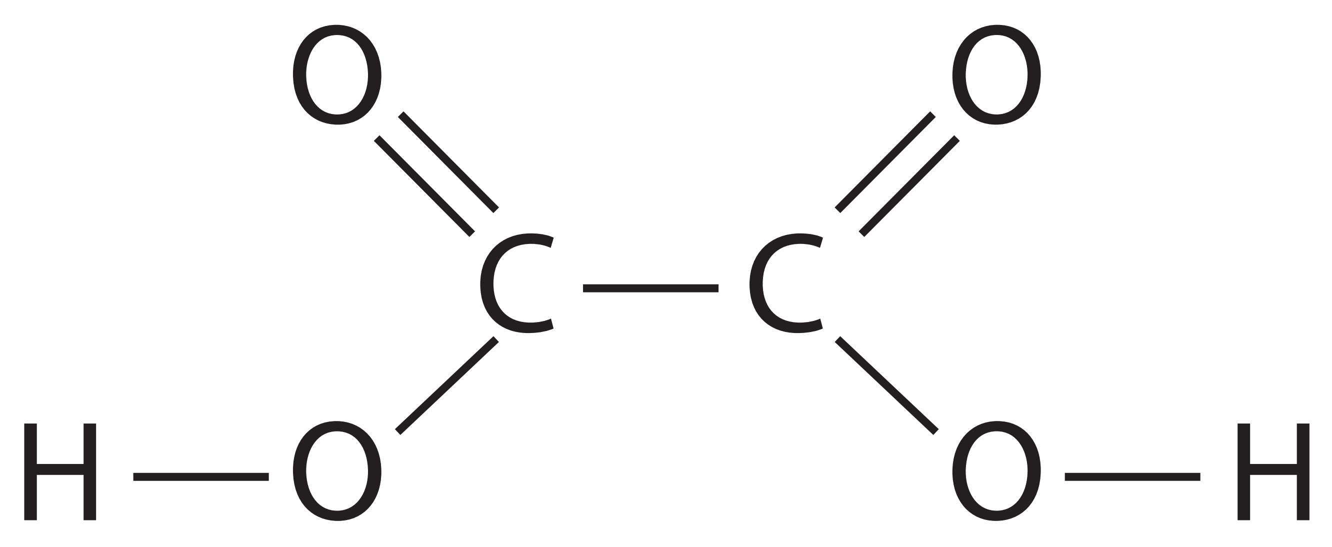
With Tollens' reagent, the silver ion present, Ag^+, is reduced to silver metal when it oxidises methanoic acid.

This oxidation of methanoic acid will also occur with stronger oxidising agents such as:

> acidified solutions of potassium manganate(VII) – decolorising the purple solution – or
>
> potassium dichromate(VI) – turning the orange solution green.

16 Oxidation of Ethandoic Acid

Another common example of a carboxylic acid that can be oxidised by these stronger oxidising agents is ethanedioic acid (a dicarboxylic acid).



ethanedioic acid

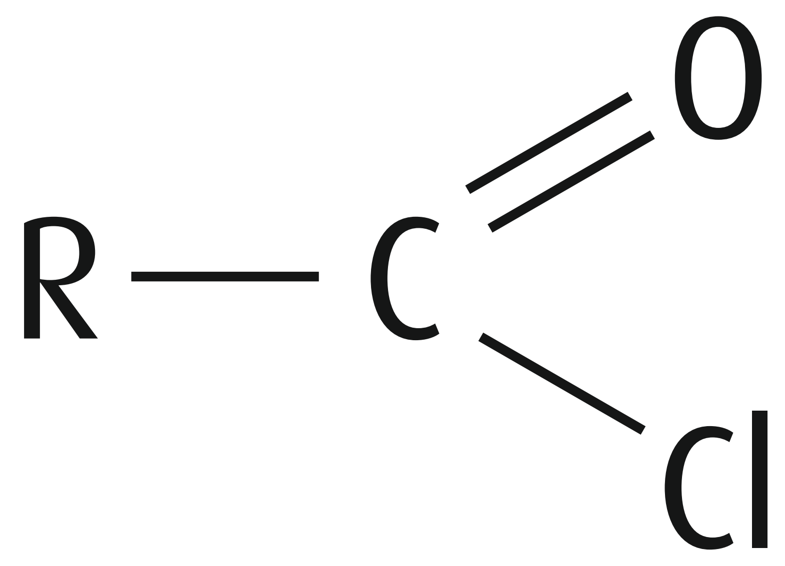
$H_2C_2O_4$

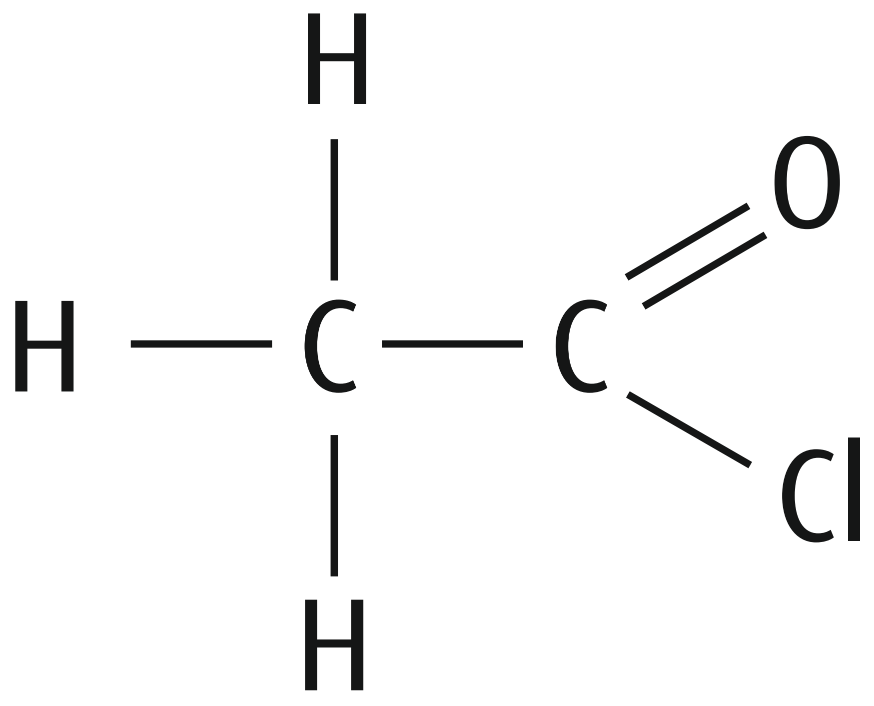
When heated with potassium manganate(VII) solution, it forms CO_2.

$$2MnO_4^- + 6H^+ + 5H_2C_2O_4 \longrightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

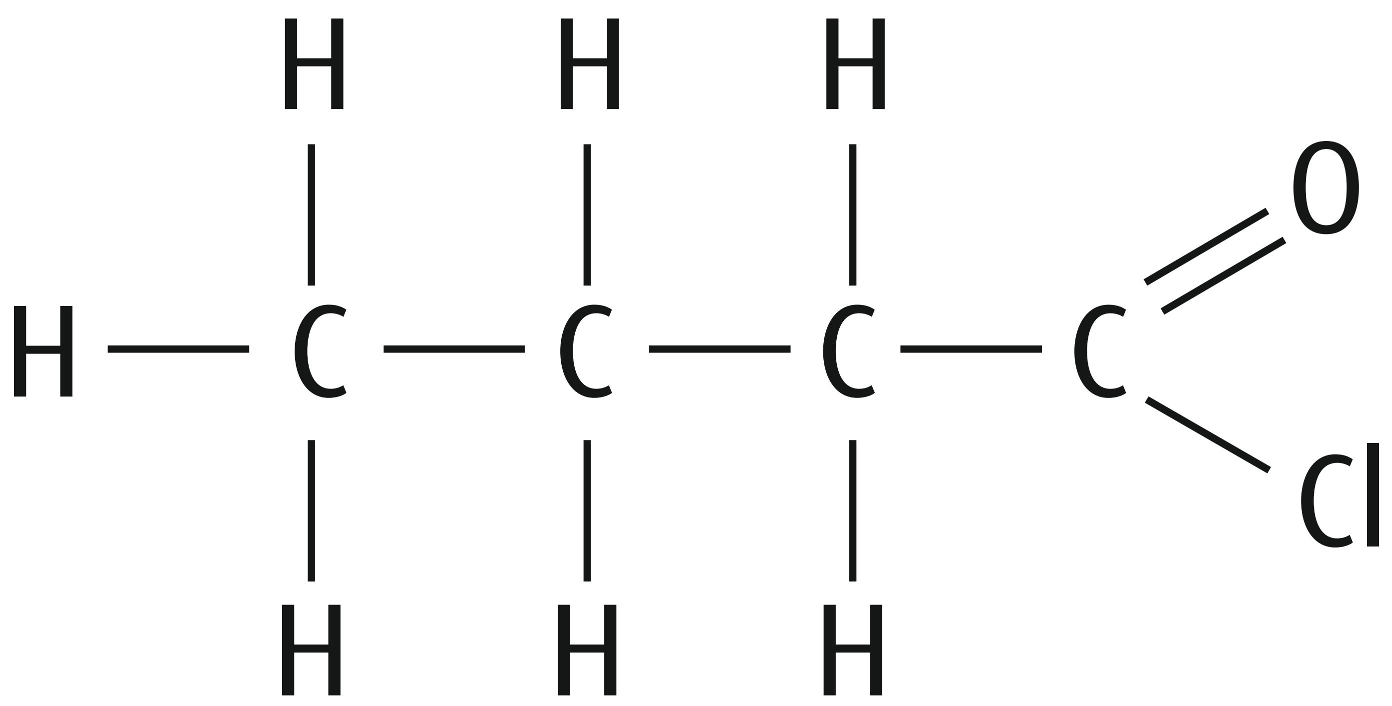
17 Acyl Chlorides

The basic structure of an acyl chloride is:

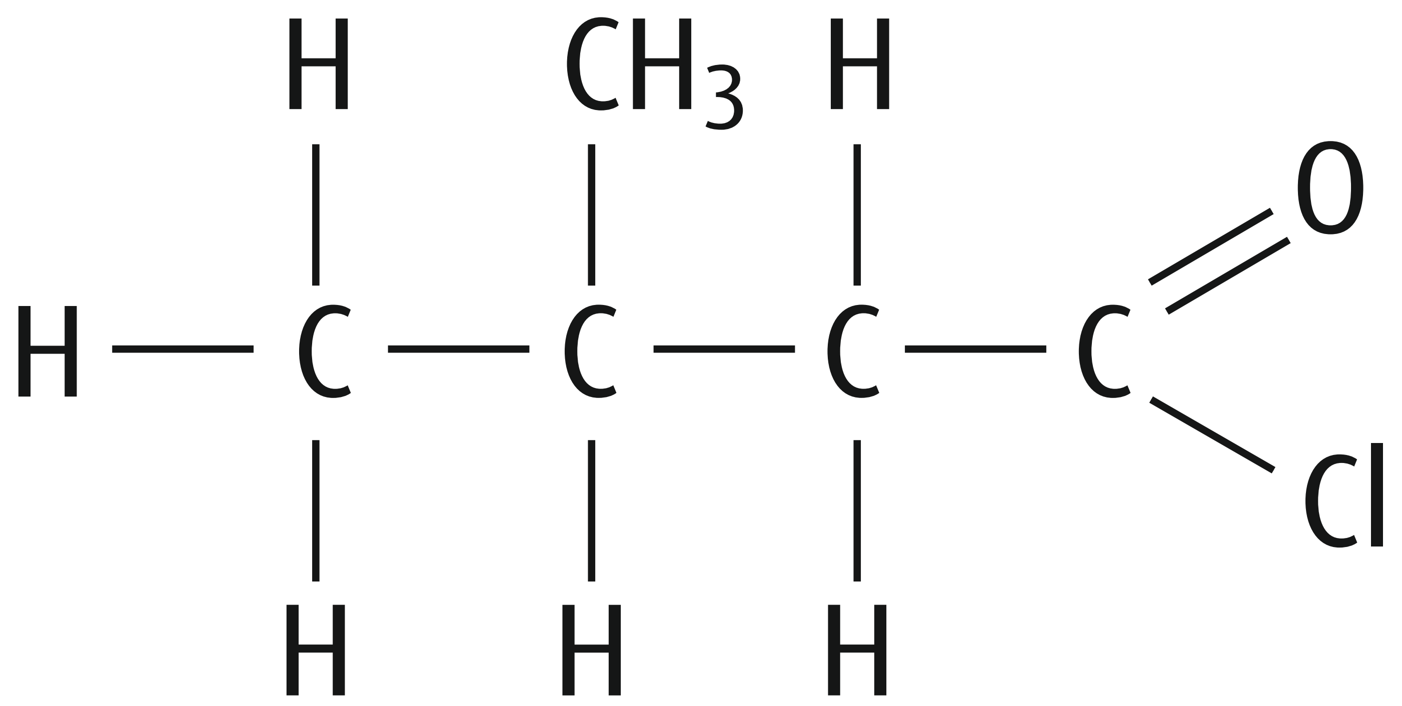




ethanoyl chloride



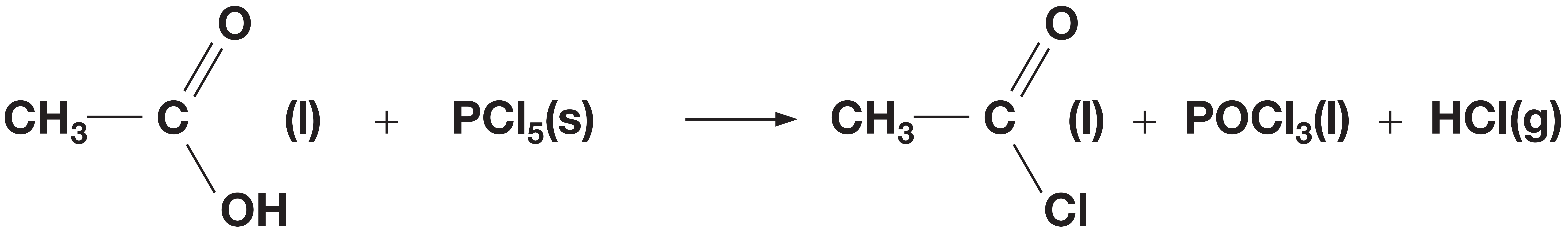
butanoyl chloride



3-methylbutanoyl chloride

18 Formation of Acyl Chlorides

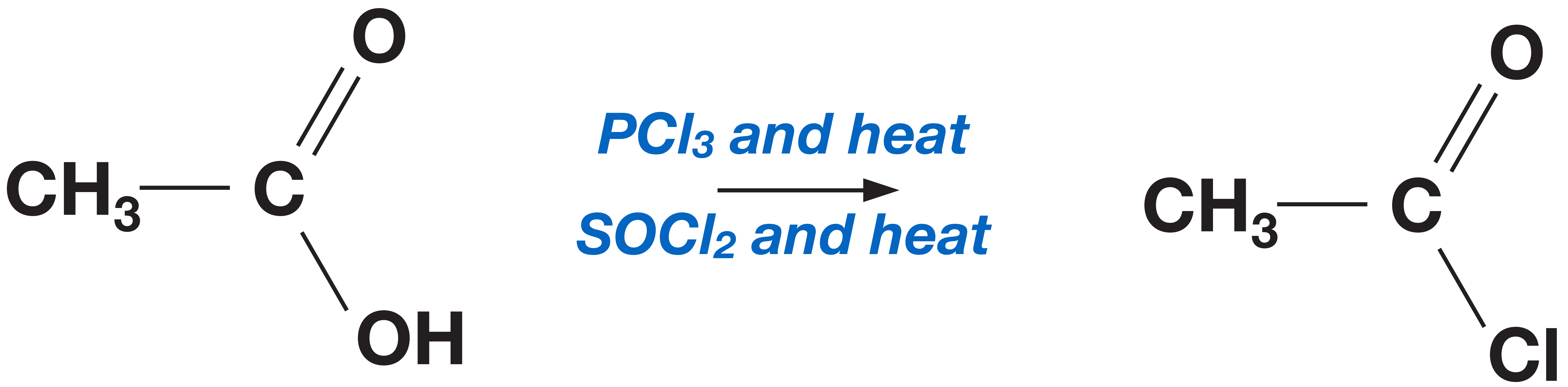
Phosphorus pentachloride, PCl_5, reacts vigorously with carboxylic acids at room temperature to form acyl chlorides. The reaction replaces the –OH group with a chlorine atom, forming an acyl chloride. The other product is hydrogen chloride gas, which fumes in moist air.



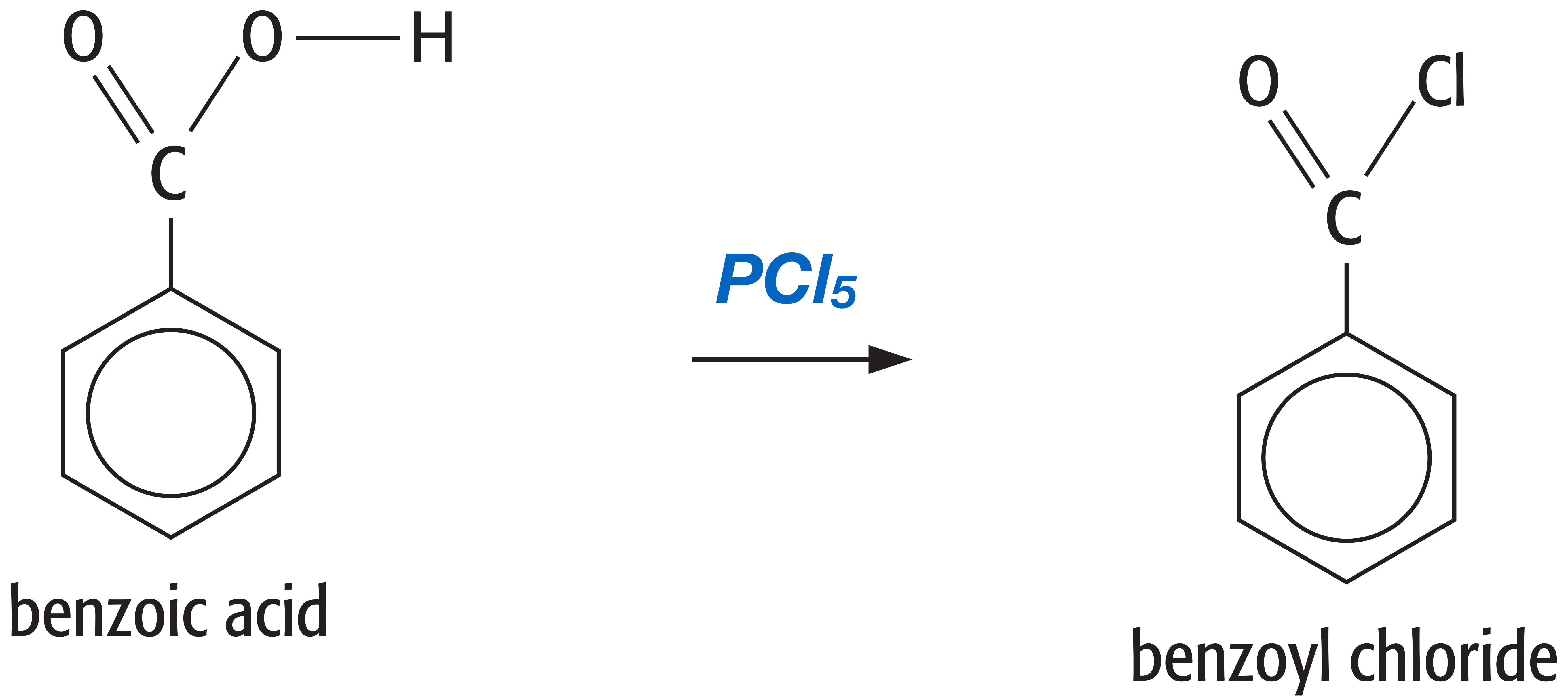
ethanoyl chloride

Formation of Acyl Chlorides

Acyl chlorides can also be made by heating carboxylic acids with phosphorus trichloride, PCl_3, or with sulfur dichloride oxide, $SOCl_2$.

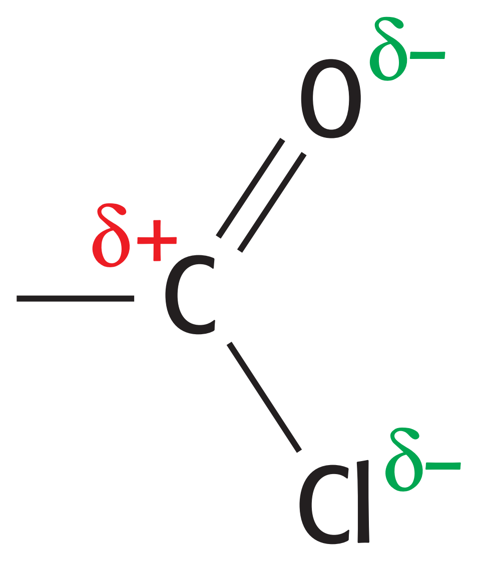


20 Formation of Acyl Chlorides



21 Reactivity

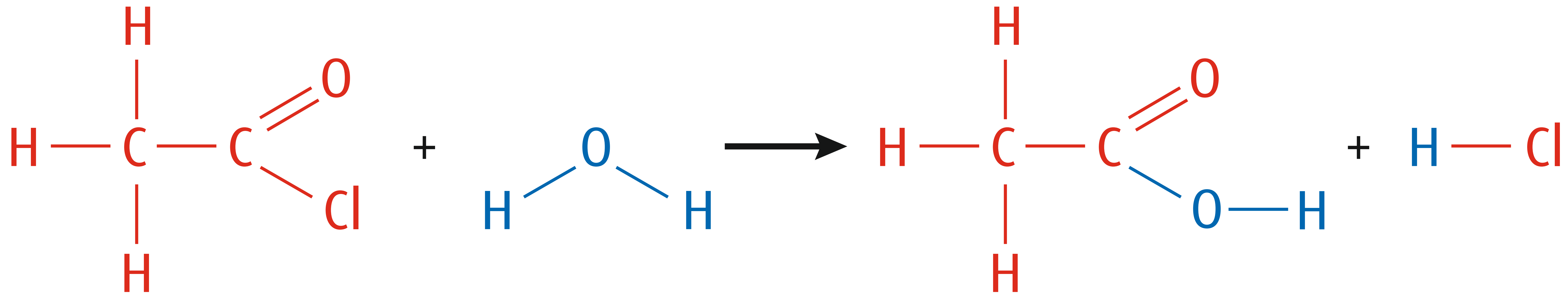
Acyl chlorides are reactive compounds. The carbonyl carbon has electrons drawn away from it by the Cl atom as well as by its O atom, and both are strongly electronegative atoms. This gives the carbonyl carbon a relatively large partial positive charge and makes it particularly open to attack from nucleophiles.



The electronegativity of the oxygen, and the easily polarised C=O double bond, have a dramatic effect on the reactivity of acyl chlorides compared with that of chloroalkanes.

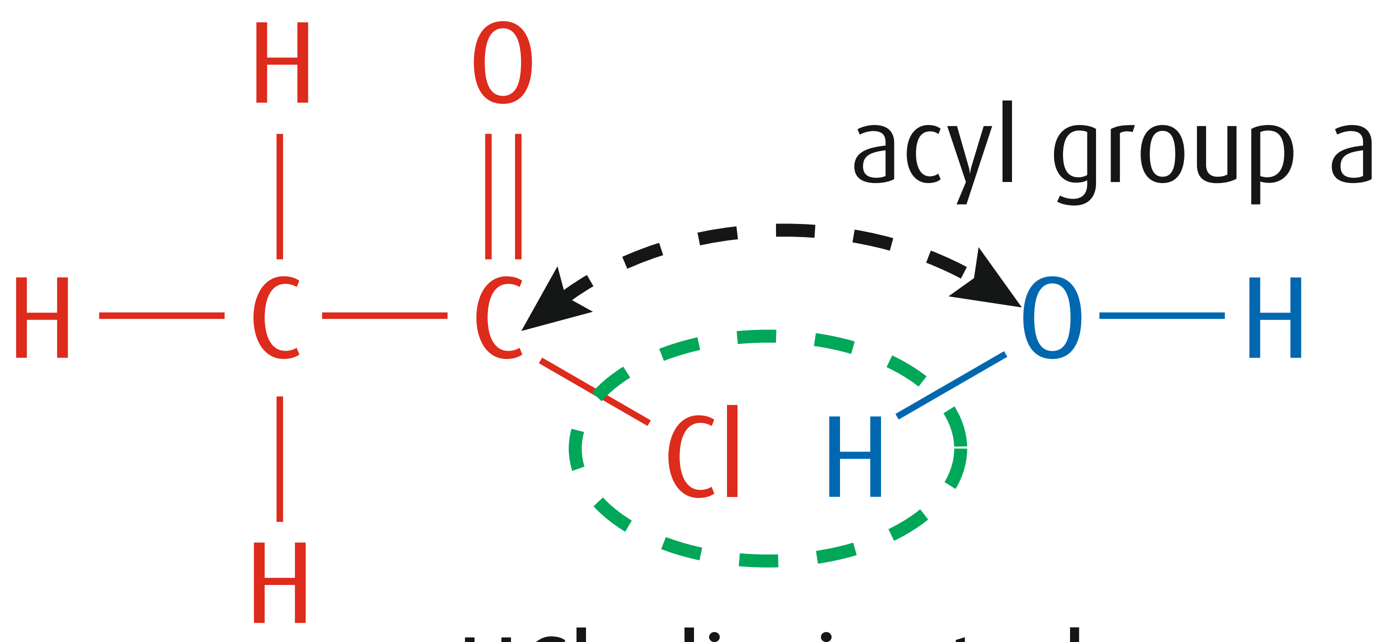
22 Reactivity

Acyl chlorides are readily hydrolysed by cold water.



ethanoyl chloride

ethanoic acid

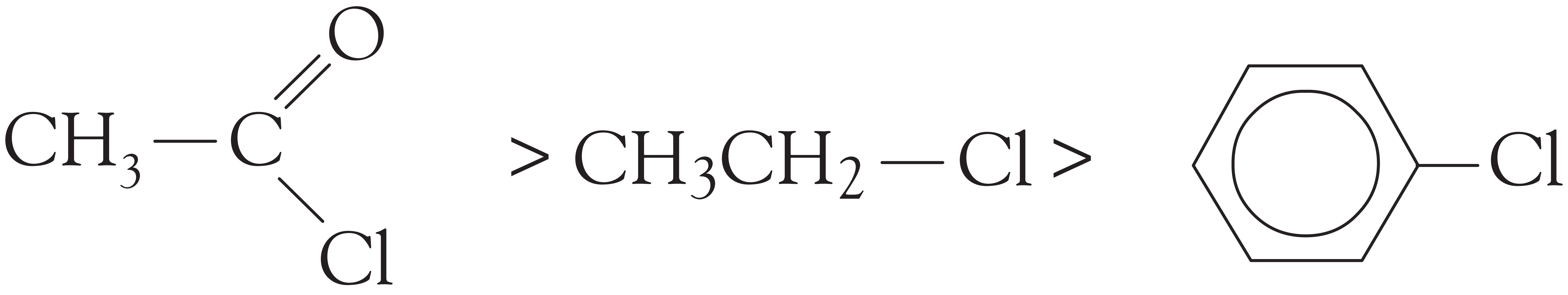


23 Reactivity

The electronegativity of the oxygen, and the easily polarised C=O double bond, have a dramatic effect on the reactivity of acyl chlorides compared with that of chloroalkanes.

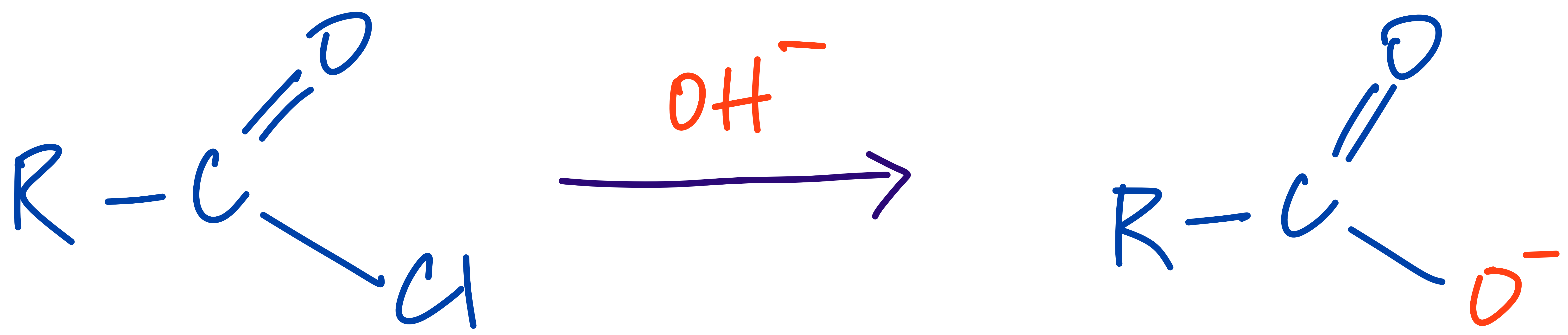
Acyl chlorides react readily with water unlike halogenalkanes and halogenobenzene.

The relative rates of hydrolysis are as follows:



24 Acyl Chlorides: Hydrolysis With Alkali

The hydrolysis of acyl chlorides is much faster with alkalis as OH^- is a stronger nucleophile than water.

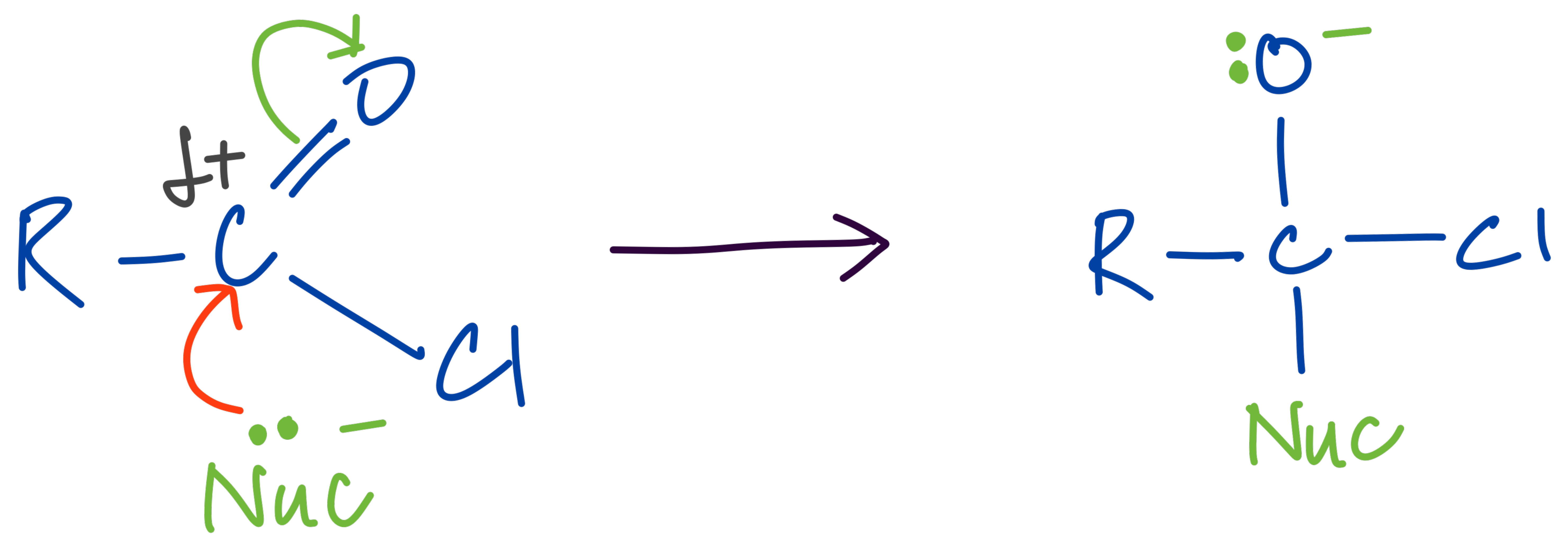


25 Addition Elimination Reaction

The carbonyl group in acyl chlorides can undergo nucleophilic addition in a similar manner to carbonyl compounds.

In the first step, the nucleophile attacks the δ^+ C of the C=O group.

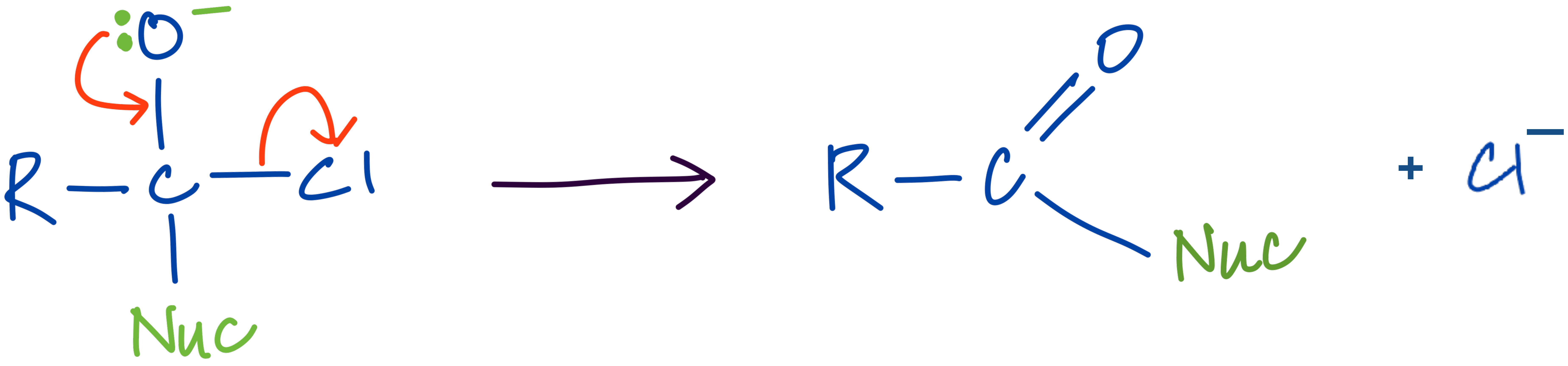
The π component of the C=O breaks with the pair of electrons going to the O to generate O^-.



26 Addition Elimination Reaction

In the second step, unlike carbonyl compounds, acyl chlorides are provided with an easily removed leaving group, the chloride ion.

A lone pair from the O then is used to re-form the double bond and at the same time the C–Cl bond breaks.

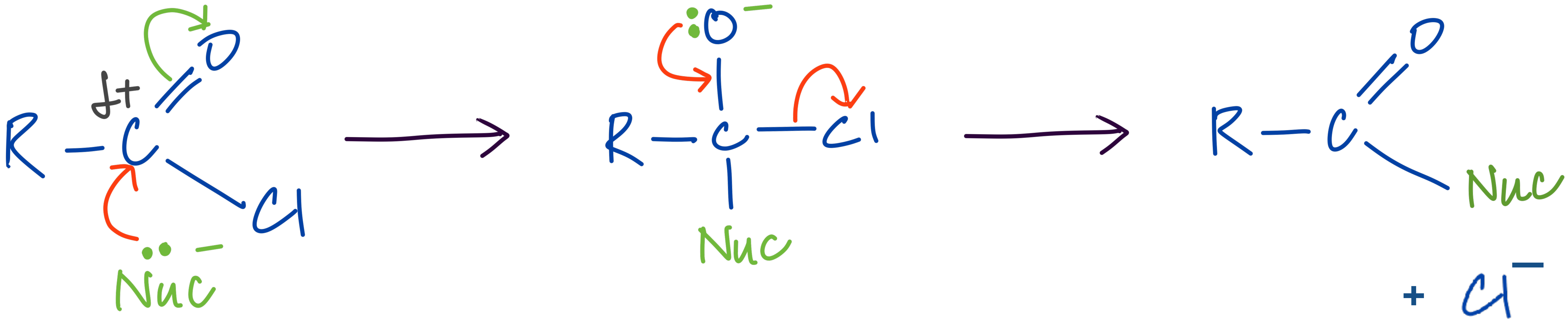


Addition Elimination Reaction

The mechanism of these reactions involves two steps:

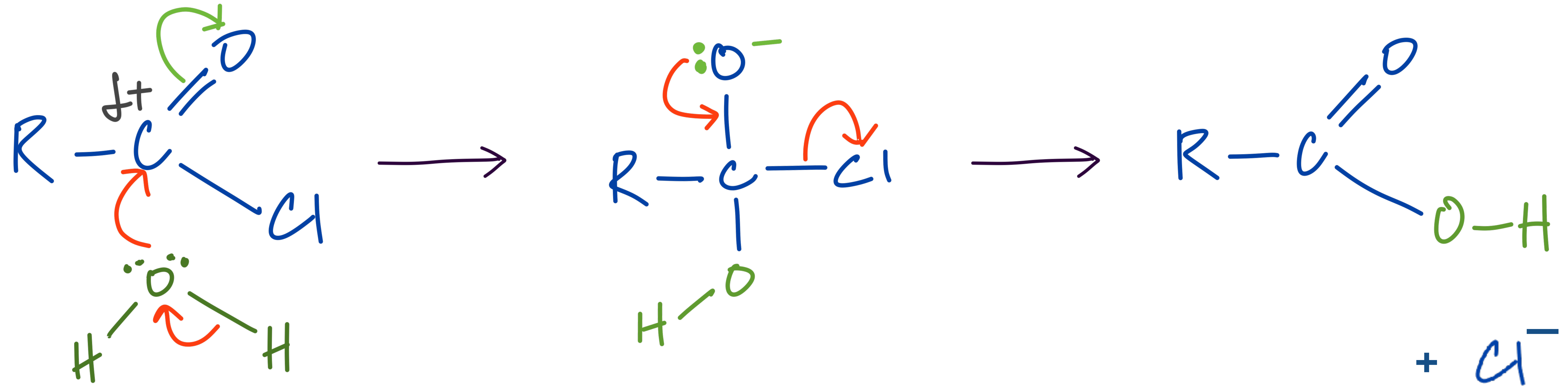
1. initial addition of the nucleophile

2. elimination of the elements of HCl.

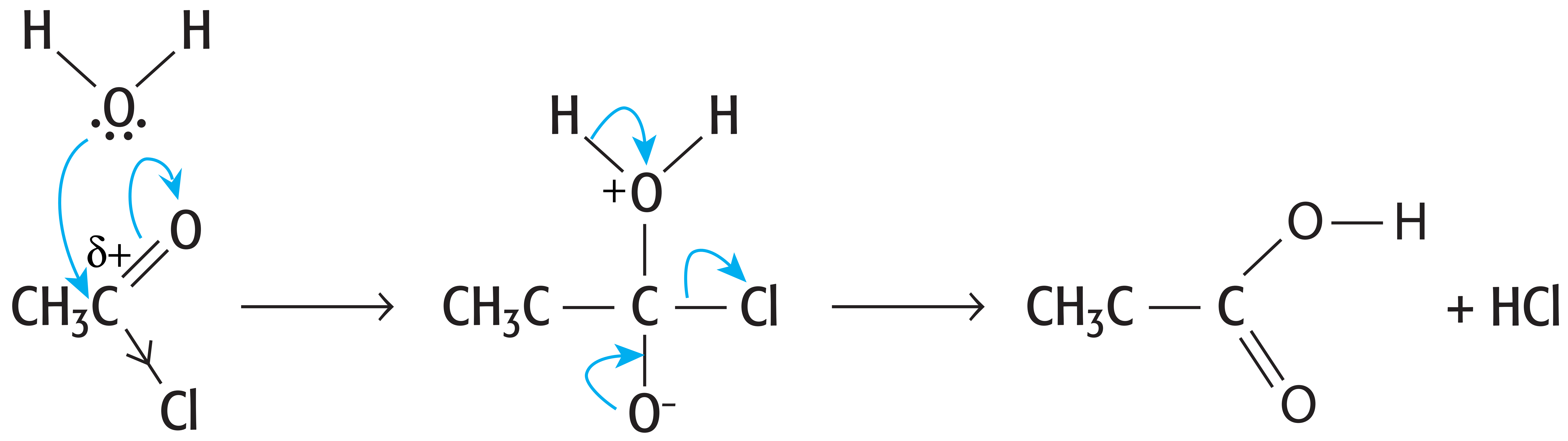


28 Addition Elimination Reaction: Hydrolysis

Nucleophilic substitution has taken place, by a mechanism involving addition, followed by elimination. If the nucleophile is water, the carboxylic acid is formed.

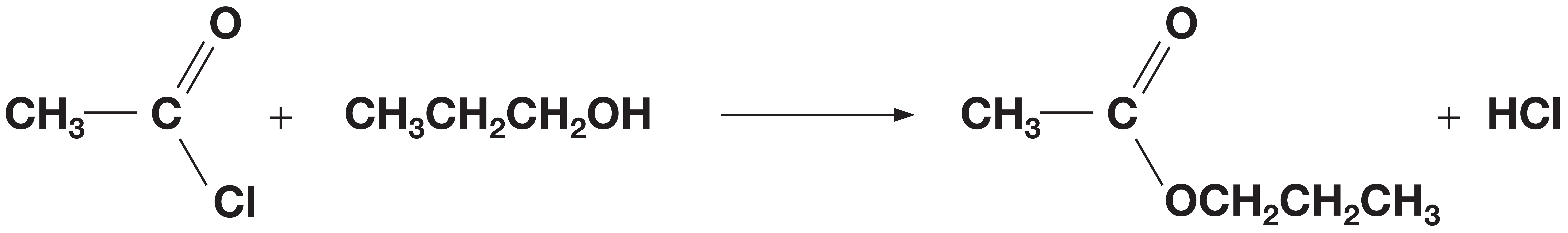


29 Addition Elimination Reaction: Hydrolysis

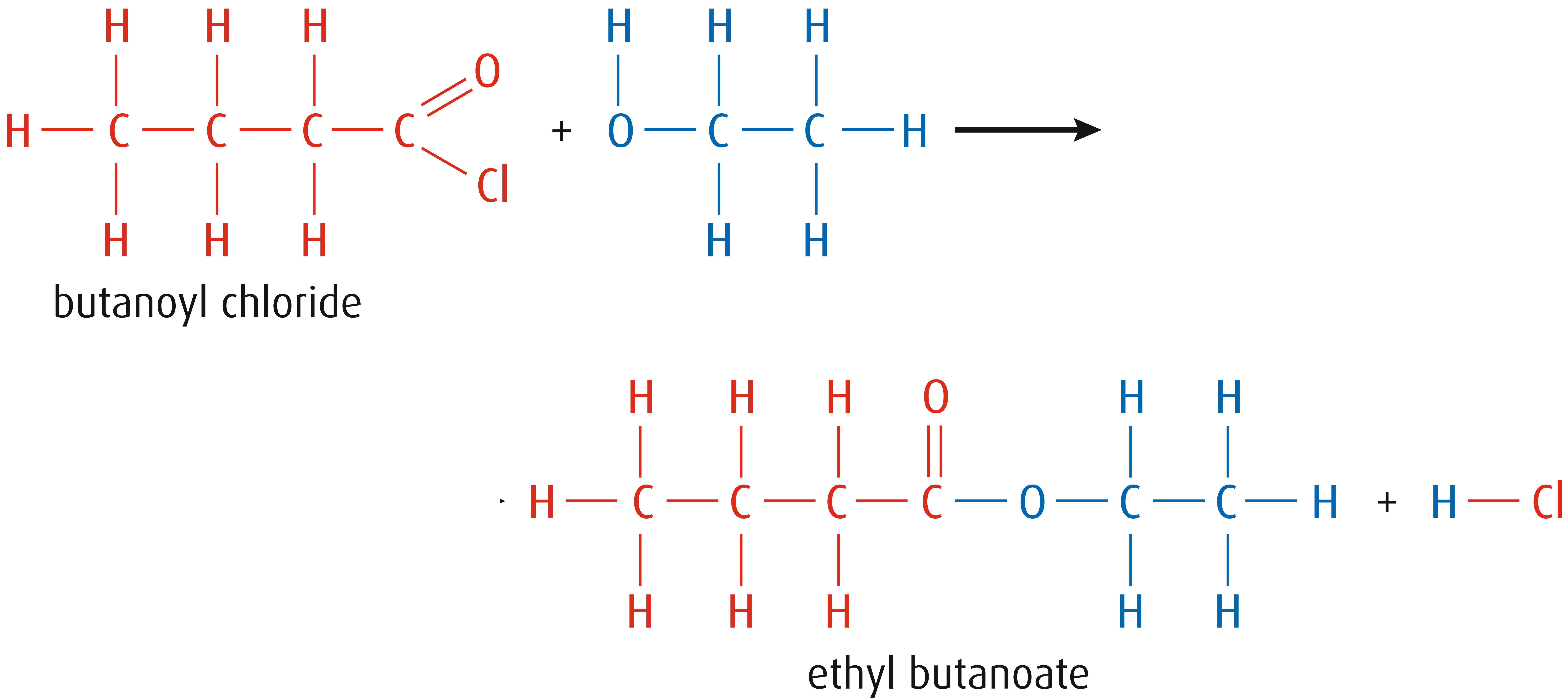


30 Acyl Chlorides With Alcohols

Acyl chlorides react with alcohols to form esters.



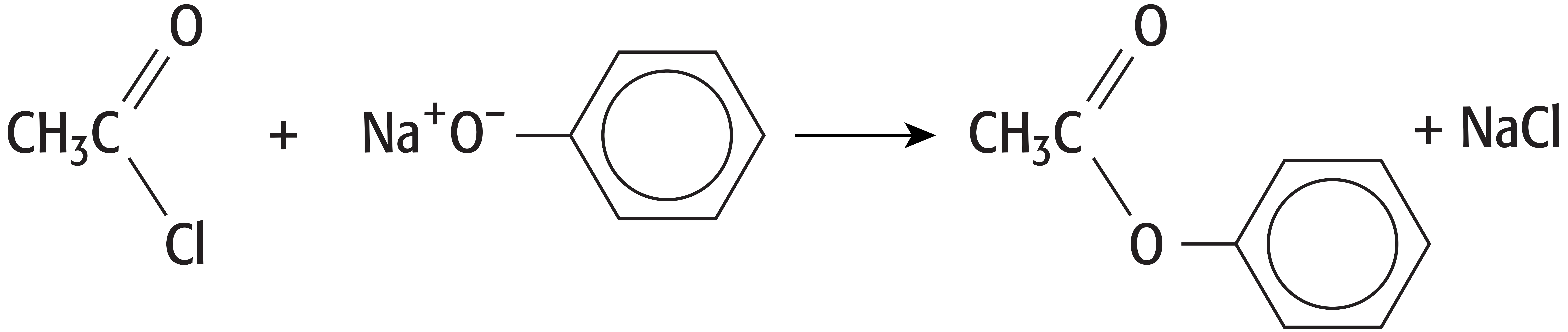
31 Acyl Chlorides With Alcohols



32 Acyl Chlorides With Phenol

An alkaline solution of phenol reacts with acyl chlorides to form phenyl esters.

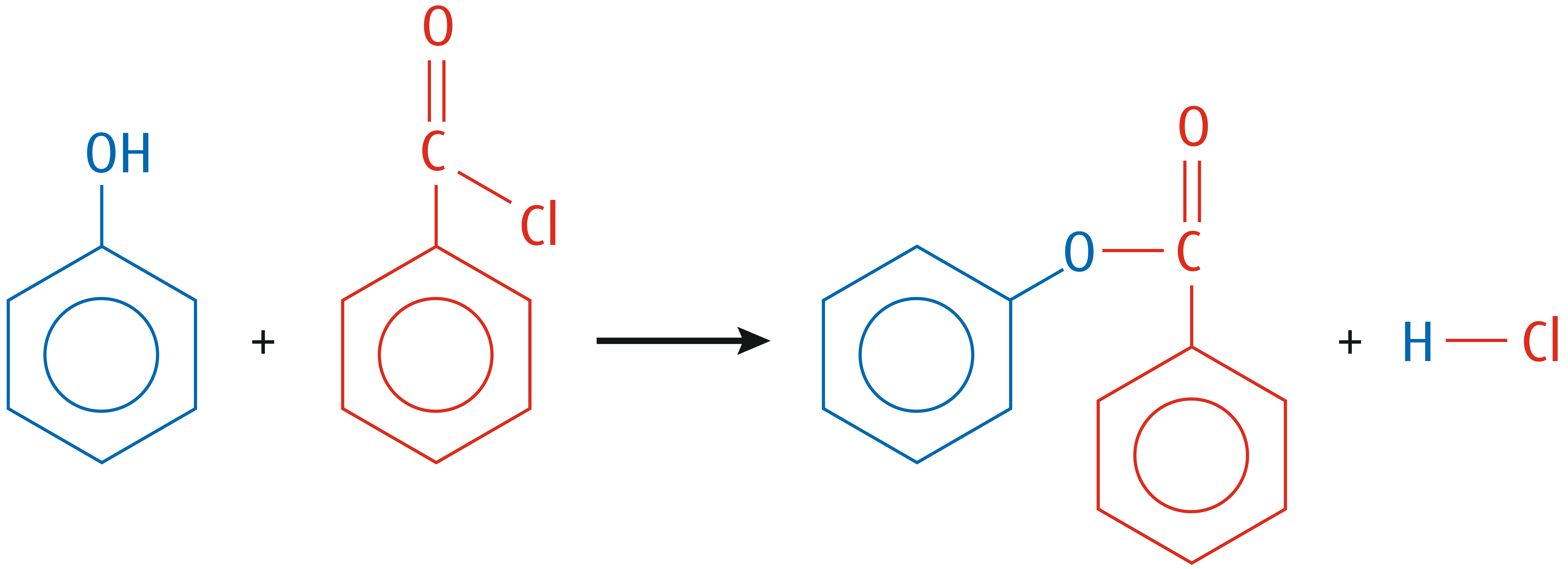
The phenol is dissolved in NaOH, to make phenoxide which is a stronger nucleophile.



phenyl ethanoate

33 Benzoyl Chloride With Phenol

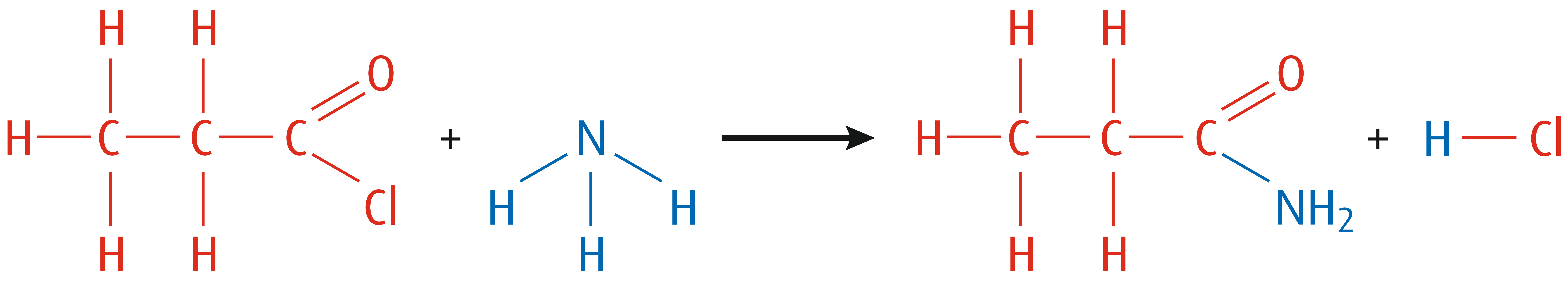
An alkaline solution of phenol reacts with benzoyl chloride to form phenyl benzoate. The phenol is dissolved in NaOH, to make phenoxide which is a stronger nucleophile.



phenyl benzoate

34 Acyl Chlorides With Ammonia

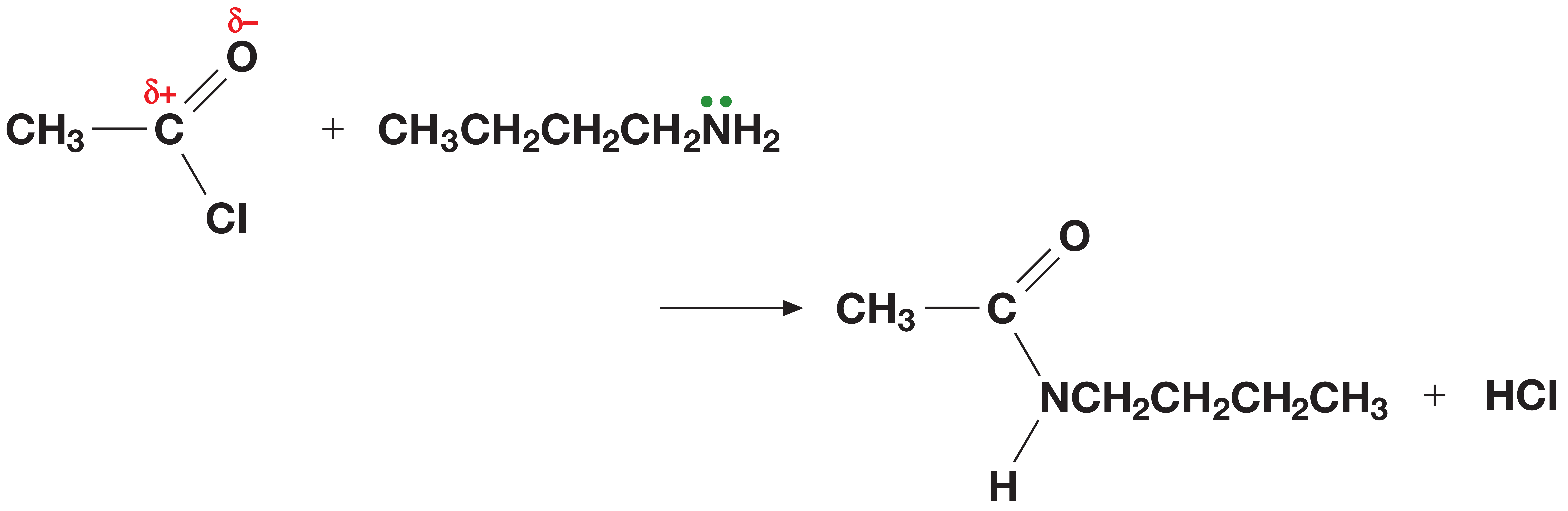
When concentrated ammonia solution is added to an acyl chloride a primary amide is formed.



propanamide

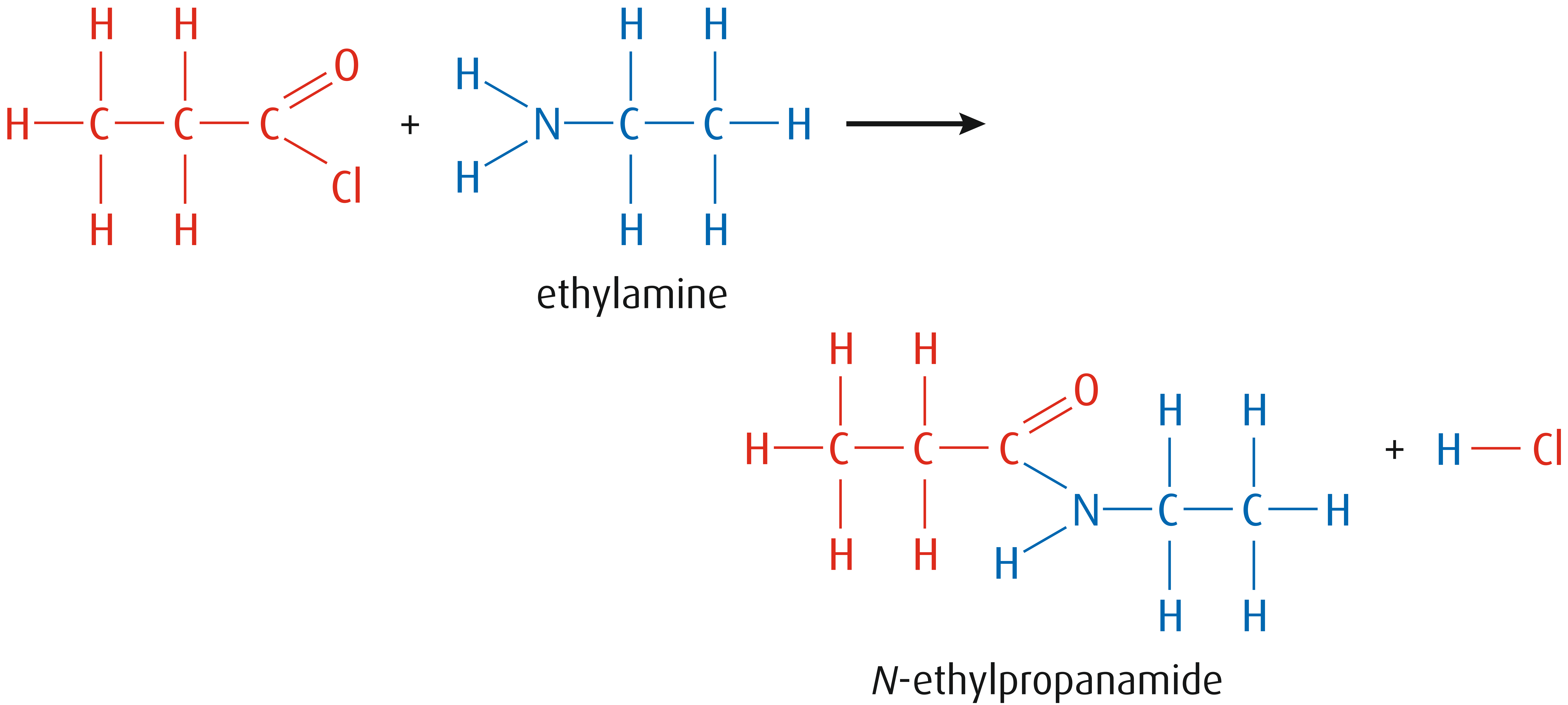
35 Acyl Chlorides With Amines

When acyl chlorides are reacted with amines N-substituted amides are formed.



N-butyl ethanamide

36 Acyl Chlorides With Amines



37 Summary of Reactions